DEPARTMENT OF PHYSICS
High Energy Physics - Weekly Seminars

مرکز پژوهشی فیزیک انرژی بالا
RESEARCH CENTER FOR HIGH ENERGY PHYSICS

The Z_n equivariant Virasoro algebra via alternative Sugawara construction

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The seminar will also be held virtually at
https://vc.sharif.edu/ch/physics-high-energy

Abstract

In this talk, I'll introduce the $u(1)^2$ Kac–Moody algebra and generalize the standard Sugawara construction of the Virasoro algebra to an infinite family of new realizations. I will show that these new realizations arise from a Z_N -grading of the mode index n of the Virasoro generators L_n . For the Z_2 and Z_3 cases, the space of all such constructions is topologically equivalent to a cylinder, while for Z_4 it forms a non-compact real four-dimensional manifold. Furthermore, I will reformulate the problem within an action principle framework by introducing Z_N -equivariant maps, which provide a systematic method for constructing conformal field theories endowed with these generalized Virasoro symmetries.